

Package #138

Implement additive manufacturing crack-resistant material inserts at rail joints to reduce hydrogen absorption and prevent fatigue fracture

Technology Maturity: ENGINEERING_DEFINITION

Transfer Posture: Sponsored Validation

Problem: Documented recurring problem 'rail steel fatigue fracture' (broken rails/joints, FRA cause family T2): 3 of 25 FRA Form 54 accidents in the retrieved window are broken-rail/joint-caused, derailing 17 cars total. Causes observed: 'Broken Rail - Base [Desc.Changed-Refer to Current/Prev.Guides]'; 'Mismatched rail-head contour'. LIMITS STAMPED INLINE: carrier-reported regulatory reports; cause codes are classifications of the initial report, NOT proven root causes; reportable-above-threshold events only (Art. XXI.5 analog).

Mechanism: Hydrogen absorption in pearlite steel (R260) during welding and operation reduces fatigue life and fracture toughness, leading to faster crack propagation and critical fracture under dynamic loads

Contents:

- 01 - Executive Technology Brief
- 02 - Engineering Technology-Transfer Dossier
- 03 - Buyer Decision Card
- 04 - Evidence Summary
- 05 - Transfer Manifest

Status: ENGINEERING_DEFINITION. Not physically validated.